

# Operations & Technician Review

Can a lower-skill employee safely deliver the right service, at the right place, and close the revenue loop?

## EXECUTIVE VERDICT

NO-GO FOR BROAD FIELD ROLLOUT. The narrow technician concept is directionally good, but assignment security, scheduling constraints, service exceptions, safety/site data, report truth, product controls, billing closeout, and durable offline operation are not yet strong enough for a low-skill workforce.

| Prepared for                                                                                               | Review date   | Scope                                                                       |
|------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------|
| Operations leadership, dispatch, customer service, field supervisors, technicians, finance, and compliance | July 15, 2026 | Business logic, operating controls, customer trust, and low-skill usability |

This review evaluates whether each feature helps the business complete a reliable customer transaction. It is intentionally not a code-quality review. The standard is operational truth: the system should guide the employee, enforce prerequisites, complete all related business effects, and visibly route exceptions.

## Review method and scope

Three independent reviews examined the product from the perspectives below, then reconciled their findings into launch gates and operating requirements. The local product source, data model, workflows, business copy, and integrations were inspected. The available hosted deployment could not be opened in the review environment, so this report does not claim a visual usability test of the running production interface.

- Operations: workload visibility, capacity, dispatch constraints, exception recovery, service-to-cash, and daily closeout.
- Technician: clarity, safety, assignment, route changes, site readiness, service evidence, products, exceptions, and offline resilience.
- Frontline control: whether a new or lower-skill employee can do the correct thing without tribal knowledge or unsafe judgment calls.

**The strongest finding is the consistency of the conclusions: leadership, customer, and field perspectives all identified the same failure class - a user can complete an apparent action while billing, scheduling, communication, or downstream records remain unfinished or contradictory.**

## What is directionally good

- The technician experience is intentionally narrow: today's route, job context, start, report, photos, GPS, and completion.
- The office day board, unscheduled pool, customer history, product catalog, payments, and recurring-job generation form a useful operations foundation.
- Finalization creates a customer-facing PDF, and recurring completion attempts to queue the next job.
- Several charge paths use idempotency protection, reducing obvious double-charge risk.
- Seven-day and one-day reminders and customer payment-method self-service can reduce office touches.

## The bulletproof technician golden path

| Step               | Required proof                                                                                         | Exception behavior                      |
|--------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 1. Download day    | Assigned route, contacts, addresses, safety/access, approved products, scope, drafts available offline | Missing data blocks dispatch            |
| 2. Navigate        | Current route and next stop acknowledged                                                               | Route change is pushed and accepted     |
| 3. Arrive          | Fresh time/location; correct address/unit; customer/access confirmation                                | No-access/wrong-site flow               |
| 4. Safety check    | Pets/people/sensitivities/do-not-treat/PPE/site conditions acknowledged                                | Unsafe condition stops work             |
| 5. Perform         | Service-specific checklist, approved product/rate/method/area, required evidence                       | Partial/deviation/supervisor flow       |
| 6. Review          | Tech sees customer-facing summary, internal notes, products, photos, charge and next consequence       | Correct before finalization             |
| 7. Complete        | Server validates assignment, state, IDs, evidence, GPS, report uniqueness, billing disposition         | Atomic/idempotent completion            |
| 8. Confirm outcome | Separate work, report, delivery, billing, and recurrence results                                       | Failed result creates owned office task |

## P0 field and operations blockers

| Priority | Business requirement                    | Acceptance condition                                                                                                                                                             |
|----------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0       | Technician identity and row scope       | The server derives tech identity from login and allows only assigned jobs/reports. Wrong links stop with a dispatch message; reassignment after start needs supervisor approval. |
| P0       | Valid job/report transitions            | Start/arrive/complete rules, timestamps, matching job/customer/tech/report IDs, one final report, and service-specific requirements are enforced server-side.                    |
| P0       | Non-completion paths                    | No access, refusal, unsafe, weather, partial, inspection-only, follow-up, callback, and incident flows determine contact, billing, reschedule, and escalation.                   |
| P0       | No lost work on technician deactivation | Deactivation blocks until future stops are resolved or atomically returns them to dispatch; no assigned job disappears from the day board.                                       |
| P0       | Safe site and scope data                | Structured unit/access/contact/safety/pest/scope/prep data is required, visible before Start, and categorized by customer/office/tech visibility.                                |
| P0       | Atomic truthful completion              | Completion is idempotent; work/report/delivery/billing/next-visit results are separate. Internal notes never appear in customer documents.                                       |
| P0       | Approved product use                    | Techs cannot activate a product. Approved label/site/pest/rate/method/PPE/certification fields and supervisor deviations are enforced.                                           |
| P0       | Service-to-cash closeout                | Every completed job has exactly one disposition: prepaid, plan-covered, charge, invoice, warranty/no-charge, or approved exception.                                              |
| P0       | Capacity-safe booking and dispatch      | Public slots are atomically reserved/revalidated; internal assignment checks duration, shift/PTO, skill, territory, equipment, drive, and windows.                               |
| P0       | Durable offline work                    | Today's data downloads, drafts autosave, actions/photos queue with stable IDs, sync status is visible, and conflicts are recoverable.                                            |
| P0       | Operations control tower                | Every critical exception has severity, owner, due time, status, resolution, and escalation.                                                                                      |
| P0       | Truthful communications                 | No screen says sent, booked, or complete when delivery or downstream effects failed; retry tasks are visible.                                                                    |

## Feature-by-feature operations disposition

| Feature                  | Verdict                | Operations requirement                                                                                                                 |
|--------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Technician login/linking | Redesign authorization | An unlinked login must not inherit a job's technician identity. Access must be row-scoped to assigned work.                            |
| My Day                   | Keep; make live        | Show route state, route changes, next action, ETA/lateness, customer contact, and missed-stop reconciliation.                          |
| Navigation               | Improve                | Provide universal map choice, verified unit/entrance, parking/access, and acknowledgement of route changes.                            |
| Route status             | Add                    | Not started, en route, in progress, and finished should reconcile automatically from job events.                                       |
| Office day board         | Keep; constrain        | Manual assignment/reorder is useful only with atomic updates, concurrent safety, unique route, and capacity checks.                    |
| Technician deactivation  | Fix before rollout     | Future assigned stops cannot remain attached to a hidden inactive technician.                                                          |
| Time windows             | Standardize            | Replace arbitrary free text with approved promised windows shared by booking, dispatch, reminders, and field screens.                  |
| Customer/site profile    | Make structured        | Unit, contact, entry, pets, sensitivities, do-not-treat, prep, prior work, exact complaint, scope, and equipment are operational data. |

| Feature                  | Verdict                    | Operations requirement                                                                                                                     |
|--------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Job scope                | Expose and require         | Initial service description and customer-critical notes must be visible; price, duration, due date, and billing disposition must be known. |
| Start job                | Add arrive/readiness       | Record en route/arrival/start timestamps and block Start until assignment, address, safety, scope, and prerequisites are valid.            |
| Job exceptions           | Build first-class flows    | Tech never improvises billability or rescheduling for no access, refusal, unsafe conditions, partial work, weather, or incident.           |
| Draft service report     | Keep; autosave             | Autosave locally and remotely, show sync, support recovery, and use service-specific structured prompts.                                   |
| GPS                      | Strengthen                 | Require fresh, accurate, radius-aware check-in/out or a documented supervisor exception; location is a flag, not an automatic accusation.  |
| Photos                   | Keep; control              | Define required before/after/condition evidence by service, compress/resume uploads, and clarify customer visibility.                      |
| Report finalization      | Rebuild transaction        | Validate IDs, assignment, state, required fields/products/GPS, one report, immutability, and idempotency server-side.                      |
| Internal notes           | Fix immediately            | Content labeled internal must never render in the customer PDF; separate visibility is a trust and HR control.                             |
| Customer report email    | Separate from completion   | Delivery failure cannot make work incomplete, but it must not display sent and must create a retry task.                                   |
| Office complete override | Manager only               | Completing without report requires reason, evidence, approval, billing choice, recurrence choice, and audit.                               |
| Product catalog          | Govern                     | Tech request -> compliance approval; include label/SDS, EPA/state, rate, method, site, PPE, re-entry, lot/expiry, cost, and inventory.     |
| One-time billing         | Close loop                 | Completion must create/trigger the selected billing outcome exactly once; dashboard surfaces completed-unbilled.                           |
| Recurring billing        | Automate and reconcile     | Successful first service triggers agreed subscription start or an urgent failed-start task.                                                |
| Recurring visit          | Anchor cadence             | Separate billing cadence, service cadence, next due window, and promised appointment; generation failure is an exception.                  |
| Reminders                | Keep; add ledger           | Deduplicate, track delivery/bounce/retry, include prep/access, and create a task when contact is required before dispatch.                 |
| Public booking capacity  | Block until atomic         | A stored quote cannot guarantee a 24-hour slot; reserve and revalidate window/skill/shift capacity at payment.                             |
| PWA/offline              | Rebuild for real field use | Caching the shell is insufficient. Data, drafts, photos, and writes must work safely during weak/no network.                               |

## Required field exception matrix

| Outcome                     | Technician must do                                   | Billing decision             | Operations result         |
|-----------------------------|------------------------------------------------------|------------------------------|---------------------------|
| No access / customer absent | Attempt contact, evidence, wait rule, dispatch alert | Approved no-access policy    | Reschedule/close decision |
| Customer refuses            | Reason and customer statement                        | No charge/fee/manager review | Account follow-up         |
| Unsafe condition            | Stop work, photo/note if safe, supervisor            | No field discretion          | Safety/incident review    |
| Weather                     | Condition and affected work                          | Company policy               | Capacity-aware reschedule |

| Outcome                | Technician must do                         | Billing decision             | Operations result                   |
|------------------------|--------------------------------------------|------------------------------|-------------------------------------|
| Partial service        | Completed/not completed scope and products | Defined partial billing rule | Follow-up work order                |
| Inspection only        | Finding and recommendation                 | Offer-specific rule          | Quote/follow-up if needed           |
| Warranty/callback      | Link original job and eligibility          | Covered/denied with approval | Quality root cause                  |
| Incident/damage/injury | Stop, secure, contact supervisor, evidence | No routine billing closeout  | Formal incident workflow            |
| Product deviation      | Stop; requested product/rate/reason        | Compliance approval          | Approved substitution or reschedule |
| GPS/no signal          | Retry and documented exception             | Supervisor override          | Flagged report, not silent bypass   |

## Minimum site-readiness record

| Category         | Required structured data                                                                                     |
|------------------|--------------------------------------------------------------------------------------------------------------|
| Location         | Validated address, property/building/unit/entrance, service areas, parking                                   |
| Access           | Occupancy, permission to enter, gate/lockbox/key/alarm, access hours, on-site contact                        |
| Safety           | Children, pets, pregnancy, respiratory/chemical sensitivity, aquariums, food handling, do-not-treat areas    |
| Problem          | Target pest, severity, exact location, observations/photos, prior treatments, warranty/callback relationship |
| Scope            | Agreed included/excluded work, service type, duration, products/method constraints, special equipment        |
| Preparation      | Customer tasks, acknowledgement, missed-prep policy, customer-visible instructions                           |
| Commercial       | Plan/one-time/warranty status, price or covered disposition, payment readiness/approved terms                |
| Notes visibility | Customer-visible, technician-critical, office-only, and internal compliance notes are separate fields        |

## Operations control tower: non-negotiable queues

| Queue              | Required exception classes                                                                                                |
|--------------------|---------------------------------------------------------------------------------------------------------------------------|
| Dispatch           | Today's unassigned; route not started; late/overdue; inactive-tech assignment; over capacity; missing readiness           |
| Customer readiness | Missing address/contact/access/prep acknowledgement; unreachable customer; requested reschedule/cancel                    |
| Service            | Past-due scheduled/in-progress; partial/no-access/unsafe; completed without final report; offline draft not synced        |
| Revenue            | Completed-unbilled; active-plan-not-billing; missing payment before visit; paid-without-invoice; failed/ambiguous payment |
| Recurrence         | Active plan without future visit; generation failed; paused/canceled plan with future jobs; cadence overdue               |
| Communications     | Reminder, report, payment, or confirmation send failed; bounce; retry exhausted; acknowledgement missing                  |
| Quality/compliance | Product approval; GPS exception; report amendment; callback; complaint; incident; license/certification gap               |

Every item needs severity, owner, due time, status, resolution reason, related customer/job, and escalation. A badge without ownership is not a control.

## P1: dependable scaled operations

- Constraint-based scheduling with shift/PTO, capacity, duration, skills/licenses, territory, equipment, time windows, drive, and overtime.
- Live route updates, arrival/start/end timestamps, acknowledgement, late/missed-stop alerts, and daily closeout.
- Service templates/checklists by pest, service, and property type; required evidence and products vary accordingly.
- Communication ledger, SMS/en-route where consented, reminder dedupe, bounce/failure recovery, and prep acknowledgement.
- Immutable reports with supervisor-approved, versioned amendments.
- A/R aging and dunning, refund/credit/void, offline-payment references, approval thresholds, and reconciliation.
- Recurring anchor/due-window/holiday/seasonality/skip rules.
- Product inventory, van stock, lot/expiry, label/SDS, certification, and replenishment.
- Duplicate customer/property detection and merge; formal incident, callback, complaint, and warranty workflows.

## P2: optimization after controls work

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- Route optimization, ETA, customer tracking, and territory profitability.
- Barcode/QR product selection and automatic inventory decrement.
- Voice-assisted structured report entry constrained to approved fields and choices.
- Technician scorecards for on-time rate, first-time completion, callbacks, report quality, safety compliance, and revenue leakage.
- Automated dispatch closeout, route reconciliation, capacity forecasts, and seasonal staffing recommendations.
- Required before/after photo prompts and image classification for quality review.

## Operations decisions engineering must not invent

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1. What outcomes count as completed and billable: treatment, inspection only, partial, no access, refusal, weather, and warranty.
2. Required fields, products, photos, checklist items, duration, and safety evidence by service type.
3. Acceptable GPS radius/accuracy/freshness and who may approve an exception.
4. Required skills, licenses, product authorizations, equipment, and supervisor rules by service.
5. When one-time billing occurs and exactly when recurring subscription billing starts.
6. What happens to existing jobs when a customer, plan, technician, or product is paused/deactivated/canceled.
7. Whether recurring cadence anchors to contract schedule or last completed service, and allowed service-window tolerance.
8. How far dispatch may move a promised date/window without customer consent and which message is required.
9. Definition and billability of callback/warranty/re-treatment work and the quality owner/SLA.
10. Which roles may add products, waive payment, complete without report, amend a final report, or override scheduling/safety rules.

## The low-skill-safe operating doctrine

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### **The system decides; the employee confirms.**

A frontline employee should see one safe next action, why it is required, and what happens next. The employee should not have to remember a hidden prerequisite, a second screen, or a cleanup step in another record.

| Principle                    | Business rule                                                                                               |
|------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>One state</b>             | Each lead, customer, plan, job, invoice, and exception has one unambiguous business state.                  |
| <b>One owner</b>             | Every open promise and exception has a named owner, due time, and escalation path.                          |
| <b>One primary action</b>    | The screen emphasizes the next valid action and suppresses invalid or premature actions.                    |
| <b>Complete transactions</b> | Activation, completion, cancellation, and payment update every affected record together.                    |
| <b>Guardrails</b>            | Price, safety, compliance, scheduling, and financial rules are enforced by the system, not training memory. |
| <b>Truthful outcomes</b>     | The app never says sent, paid, booked, active, or complete unless that exact result is durable.             |
| <b>Manager exceptions</b>    | Overrides require a role, reason, evidence where appropriate, and an immutable audit event.                 |
| <b>Recoverable failure</b>   | A partial failure creates a visible work item; it never disappears into a log or a green success screen.    |

## Selected evidence

| Finding                                     | Representative source evidence                                                                       |
|---------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Tech access / identity</b>               | apps/web/amplify/data/resource.ts:339-360, 391-395, 437-470; apps/crm/src/tech/JobDetail.tsx:101-105 |
| <b>Schedule and lost inactive-tech work</b> | apps/crm/src/office/Schedule.tsx:47-67, 79-158, 242-289, 433-455                                     |
| <b>Missing field scope/site data</b>        | agreement-public/handler.ts:110-123; crm/src/tech/JobDetail.tsx:137-170                              |
| <b>Job states/start</b>                     | data/resource.ts job model; JobDetail.tsx:173-184                                                    |
| <b>Finalization controls</b>                | JobDetail.tsx:483-509; crm-docs/handler.ts:190-239, 261-309                                          |
| <b>Internal notes exposure</b>              | JobDetail.tsx internal-notes label; crm-docs PDF generation and technician notes rendering           |
| <b>Products</b>                             | JobDetail.tsx:556-574; data/resource.ts:424-440                                                      |
| <b>GPS</b>                                  | JobDetail.tsx:262-288; crm-docs/handler.ts finalization validation                                   |
| <b>Offline</b>                              | apps/crm/public/sw.js:25-128                                                                         |
| <b>Billing closeout</b>                     | crm-billing/handler.ts:272-335; Dashboard.tsx:88-190; shared/bookingFinalize.ts:136-203              |
| <b>Recurring</b>                            | shared/recurring.ts:4-27, 83-111                                                                     |
| <b>Booking capacity</b>                     | booking-public/handler.ts:527-547; booking-public/availability.ts:138-193                            |
| <b>Payment CTA</b>                          | crm-docs/handler.ts:118-125; crm/src/App.tsx:206-213                                                 |

Evidence paths are relative to the BuzzKill repository. They illustrate the business behavior supporting the operations requirements.